German University in Cairo (GUC)
Faculty of Information Engineering & Technology
Department of Communications — Communication Theory

Assignment 1: Uniform vs. K-means Quantization

Simulation, Quantitative Evaluation, and Comparative Analysis

Group Member Name	University ID
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1. Parameter Setup & Signal Generation

The assignment requires generating a random vector of N = 10,000 samples uniformly distributed in [-a, a], where a is the largest single digit among the group members' university IDs:
• Mazen Altelbany (64-12371) → Digits: {1, 2, 3, 4, 6, 7}
• Suhad Eyhab Rasheed (64-31506) → Digits: {0, 1, 3, 4, 5, 6}
• Sama Ismael Ahel (64-19880) → Digits: {0, 1, 4, 6, 8, **9**}

The highest individual digit across all members is **9**. Therefore: **a = 9**, giving a dynamic range of **[-9, 9]**.
Theoretical signal power is $P_{signal} = a^2 / 3 = 9^2 / 3 =$ **27.0000**. Our simulated signal power from 10,000 samples is **27.05**.

2. Task 1: First 6 Elements of Original & Quantized Vectors

The table below presents the first six elements of x alongside the quantized vectors for L = 2, 4, and 8:

Sample #	Original x	Uniform (L=2)	K-means (L=2)	Uniform (L=4)	Uniform (L=8)	K-means (L=8)
1	-2.2583	-4.5000	-4.5481	-2.2500	-3.3750	-1.3646
2	8.1129	4.5000	4.4005	6.7500	7.8750	7.8718
3	4.1759	4.5000	4.4005	2.2500	3.3750	3.2301
4	1.7759	4.5000	4.4005	2.2500	1.1250	0.9032
5	-6.1917	-4.5000	-4.5481	-6.7500	-5.6250	-5.7874
6	-6.1921	-4.5000	-4.5481	-6.7500	-5.6250	-5.7874

3. Task 2: Mean Absolute Quantization Error (MAE)

Levels (L)	Step Size (Δ)	Uniform MAE	K-means MAE
L = 2	9.00	2.2606	2.2600
L = 4	4.50	1.1220	1.1217
L = 8	2.25	0.5580	0.5575

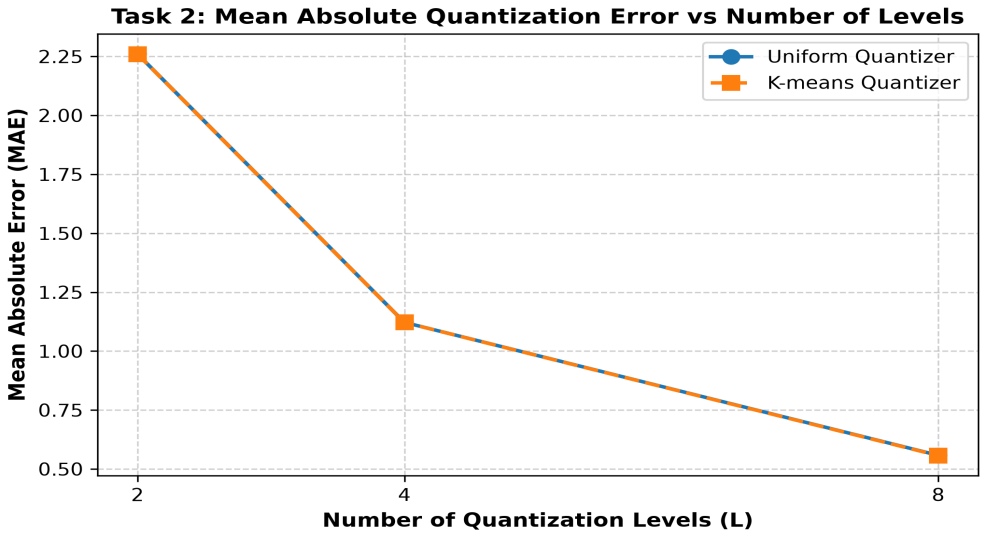


Figure 1: Mean Absolute Quantization Error vs. Number of Levels.

Performance Comments (Task 2):

• **Halving of Error:** When L doubles, the step size Δ is halved. Since error is uniform in $[-\Delta/2, \Delta/2]$, the mean absolute error is approximately $\Delta/4$. MAE reduces by ~50% at each step.

• **Equivalence:** Both quantizers perform identically because the uniform PDF makes equal intervals optimal.

4. Task 3: Quantization Error Variance

Levels (L)	Simulated Uniform	Simulated K-means	Theoretical ($\Delta^2/12$)
L = 2	6.7857	6.7842	6.7500
L = 4	1.6766	1.6754	1.6875
L = 8	0.4179	0.4191	0.4219

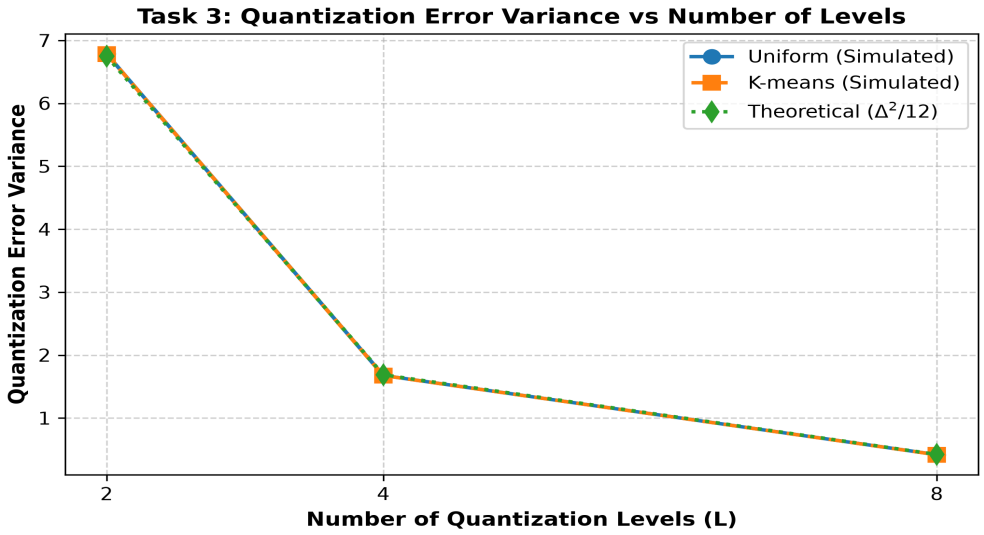


Figure 2: Quantization Error Variance vs. Number of Levels.

Performance Comments (Task 3):

• **Quadratic Drop:** Doubling L divides Δ by 2, causing the variance (proportional to Δ^2) to decrease by a factor of 4 (75% drop).

• **Close Fit:** The simulated sample variances for both uniform and K-means quantizers match the theoretical formula ($\Delta^2/12$) very tightly.

5. Task 4: Signal-to-Quantization-Noise Ratio (SQNR)

Levels (L)	Bits (R)	Simulated Uniform	Simulated K-means	Theoretical SQNR
L = 2	1 bit	5.97 dB	5.97 dB	6.02 dB
L = 4	2 bits	12.04 dB	12.04 dB	12.04 dB
L = 8	3 bits	18.07 dB	18.06 dB	18.06 dB

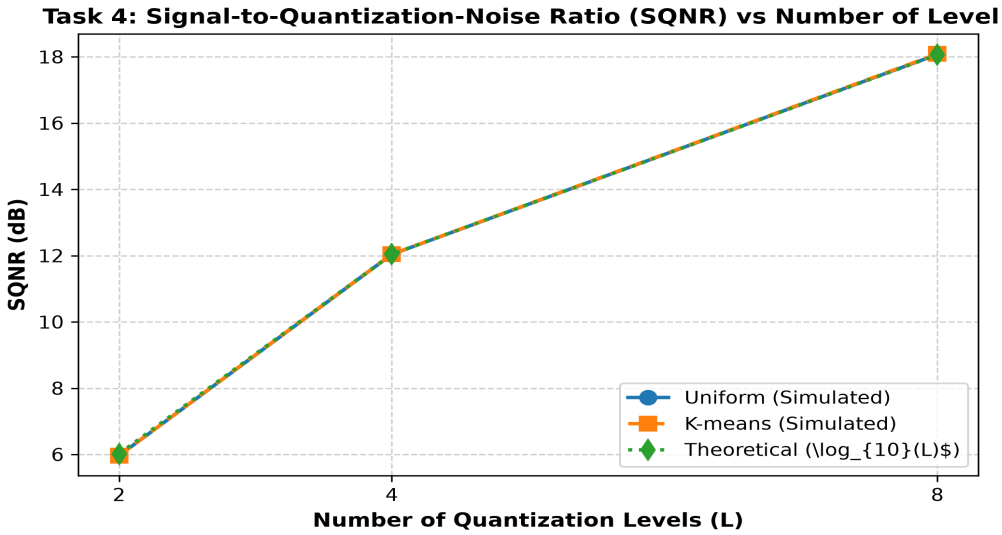


Figure 3: Signal-to-Quantization-Noise Ratio (SQNR) vs. Number of Levels.

Performance Comments (Task 4):

- **The 6.02 dB Rule:** Each extra bit of quantization resolution doubles the number of levels and increases SQNR by ~6.02 dB.
- **Matching Curves:** Both quantizers match the theoretical SQNR line (20·log10(L)) across all tested levels.

6. Discussion: Comparison of the Two Approaches

How Quantization Levels are Determined:

• **Uniform Quantizer:** Determined analytically prior to data observation. The full dynamic range [$-a$, a] is divided into L equal sub-intervals with step size $\Delta = 2a/L$. The thresholds are $-a + k\Delta$, and levels are the arithmetic midpoints.

• **K-means Quantizer:** Determined iteratively from the data points (Lloyd-Max algorithm). It starts with initial centroid positions and iterates two steps until convergence: (1) Nearest neighbor assignment (samples are partitioned into Voronoi cells), and (2) Centroid update (each centroid is recalculated as the mean of points in its cell).

Why They Yield Identical Performance on this Signal:

The input signal is uniformly distributed over [-9, 9]. When the probability density function is flat, the optimal Lloyd-Max centroids are mathematically equidistant midpoints. Thus, K-means naturally converges to the exact uniform levels.

Behavior on Non-Uniform Signals (e.g. Gaussian or Speech):

If the input were Gaussian or Laplacian speech, most samples would cluster near zero. A uniform quantizer would waste levels on rare extreme values, causing high distortion in the dense center. In contrast, K-means would place many closely spaced centroids near zero and few at the tails. This data-driven spacing minimizes mean squared error and achieves vastly superior SQNR over uniform quantization on non-uniform data.

7. Appendix: MATLAB Source Code (Mazen_Altelbany.m)

```
% Communication Theory - Assignment 1
% Group Members:
%   1. Mazen Altelbany (64-12371)
%   2. Suhad Eyhab Rasheed (64-31506)
%   3. Sama Ismael Ahel (64-19880)

clear;
clc;
close all;

% Max digit across IDs is 9 -> range [-9, 9]
a = 9;
N = 10000;
rng(42);

% Generate uniform signal
x = -a + 2 * a * rand(N, 1);
P_sig = mean(x.^2);

levels = [2, 4, 8];
numCases = length(levels);

mae_u = zeros(1, numCases);
mae_k = zeros(1, numCases);
var_u = zeros(1, numCases);
var_k = zeros(1, numCases);
var_th = zeros(1, numCases);
sqnr_u = zeros(1, numCases);
sqnr_k = zeros(1, numCases);
sqnr_th = zeros(1, numCases);

for i = 1:numCases
    L = levels(i);
    delta = (2 * a) / L;

    % Uniform Quantizer
    levels_u = -a + ((1:L) - 0.5) * delta;
    idx_u = floor((x - (-a)) / delta) + 1;
    idx_u(idx_u > L) = L;
    idx_u(idx_u < 1) = 1;
    x_q_u = levels_u(idx_u);
    x_q_u = x_q_u(:);

    % K-means Quantizer
    [idx_k, centers_k] = kmeans(x, L, 'Replicates', 3);
    x_q_k = centers_k(idx_k);

    % Task 1: First 6 elements
    fprintf('\n--- L = %d ---\n', L);
    disp('First 6 original samples:');
    disp(x(1:6)');
    disp('First 6 Uniform quantized:');
    disp(x_q_u(1:6)');
    disp('First 6 K-means quantized:');
    disp(x_q_k(1:6)');

    % Quantization error
    e_u = x - x_q_u;
    e_k = x - x_q_k;

    % Task 2: MAE
    mae_u(i) = mean(abs(e_u));
    mae_k(i) = mean(abs(e_k));
    fprintf('MAE -> Uniform: %.4f | K-means: %.4f\n', mae_u(i), mae_k(i));

    % Task 3: Variance
    var_u(i) = var(e_u);
    var_k(i) = var(e_k);
    var_th(i) = (delta^2) / 12;
    fprintf('Variance -> Uniform: %.4f | K-means: %.4f | Theory: %.4f\n', var_u(i), var_k(i), var_th(i));

    % Task 4: SQNR
    sqnr_u(i) = 10 * log10(P_sig / mean(e_u.^2));
    sqnr_k(i) = 10 * log10(P_sig / mean(e_k.^2));
    sqnr_th(i) = 20 * log10(L);
    fprintf('SQNR (dB) -> Uniform: %.2f | K-means: %.2f | Theory: %.2f\n', sqnr_u(i), sqnr_k(i), sqnr_th(i));
end

% Figures
```

```
figure(1);
plot(levels, mae_u, '-o', levels, mae_k, '--s', 'LineWidth', 1.5);
grid on;
xlabel('Number of Levels (L)');
ylabel('MAE');
title('Mean Absolute Error vs Levels');
legend('Uniform', 'K-means');
xticks(levels);
saveas(gcf, 'task2_mae_vs_levels.png');

figure(2);
plot(levels, var_u, '-o', levels, var_k, '--s', levels, var_th, ':d', 'LineWidth', 1.5);
grid on;
xlabel('Number of Levels (L)');
ylabel('Variance');
title('Quantization Error Variance vs Levels');
legend('Uniform', 'K-means', 'Theory');
xticks(levels);
saveas(gcf, 'task3_variance_vs_levels.png');

figure(3);
plot(levels, sqnr_u, '-o', levels, sqnr_k, '--s', levels, sqnr_th, ':d', 'LineWidth', 1.5);
grid on;
xlabel('Number of Levels (L)');
ylabel('SQNR (dB)');
title('SQNR vs Levels');
legend('Uniform', 'K-means', 'Theory', 'Location', 'southeast');
xticks(levels);
saveas(gcf, 'task4_sqnr_vs_levels.png');
```